

COMIC BOOKS DATASET

Exploratory Data Analysis Report

Global Comic Industry — Market Intelligence & Audience Insights

10,000+ Total Comics Dataset size	5,005 Completed Titles 49% of catalogue	8.06 / 10 Avg Rating All age groups	2023 Peak Year ~595 releases
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Dataset: Comic Books Dataset.csv
Tools: Python · Pandas · Plotly · Seaborn · Matplotlib

Executive Summary

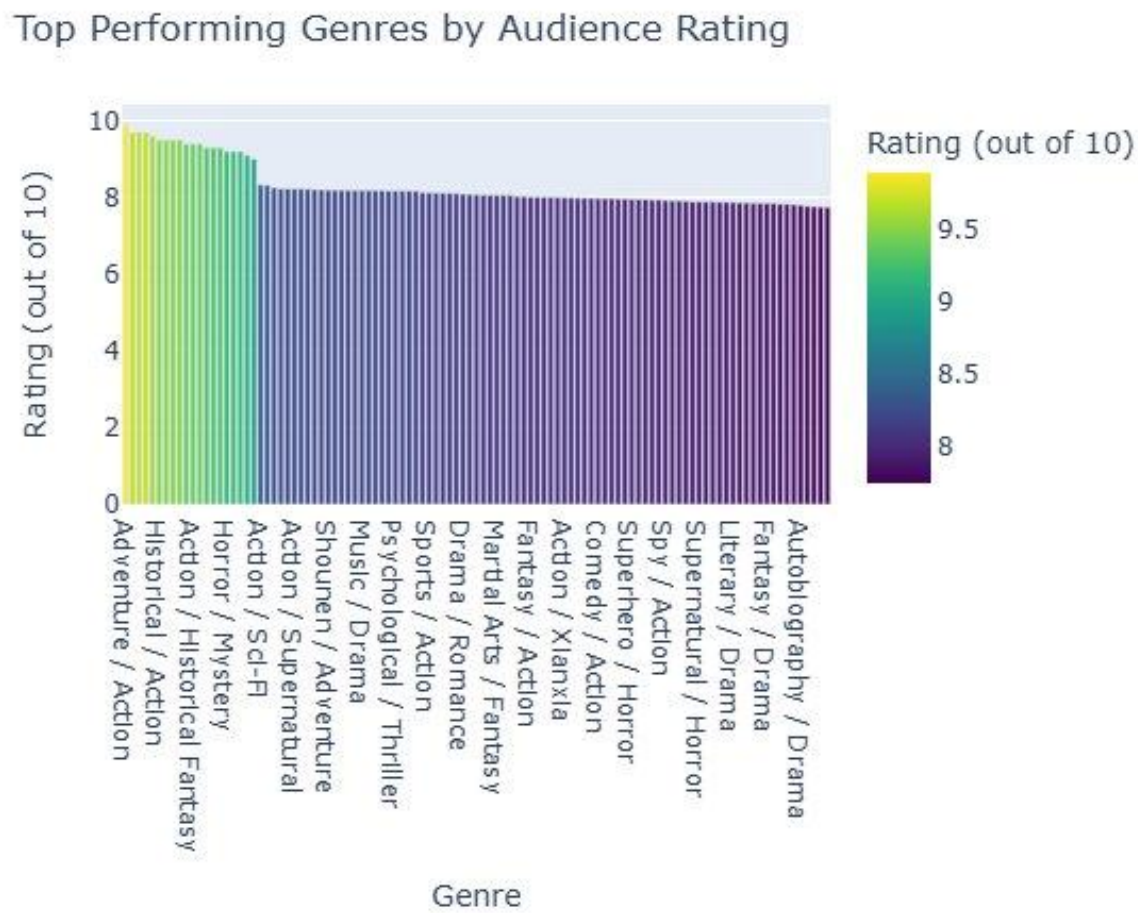
This report presents a comprehensive exploratory data analysis (EDA) of the Global Comic Books Dataset. The analysis examines over 10,000 comic titles spanning genres, formats, countries of origin, age categories, visual styles, and business metrics — delivering actionable insights for publishers, platform operators, and content strategists.

Key findings reveal that Action-hybrid genres consistently dominate audience satisfaction, the global comic industry peaked in 2023 before a sharp correction, and Japanese formats hold the largest collective market share. Audience ratings are remarkably uniform across age groups and visual styles, indicating that story quality and execution — not classification or format — are the primary drivers of satisfaction.

Dimension	Key Finding
Genre Performance	Adventure/Action leads at ~9.8/10; Action-hybrid genres sweep the top 4
Production Trend	Industry peaked in 2023 (~595 titles); 2026 shows sharp correction to ~450
Age Ratings	All five age groups score within a 0.038-point band (8.04–8.08)
Country of Origin	France/Iran and France/Spain top the charts; East Asia dominates mid-tier
Format Distribution	No single format commands a majority; Japanese formats lead collectively at 27%
Color Style Impact	Medians are identical (~8.0) across all 8 visual styles
Genre-Country Mix	Superhero/Thriller leads globally at 94 count; Superhero genres hold 8 of top 30 slots
Business Correlations	Max correlation is 0.42 (Page Count vs Volume Count); ratings are independent of all metrics
Page Count vs Ratings	Quality is format-agnostic; top ratings appear at all page counts
Production Pipeline	Completed:Cancelled ratio is 11.5:1, reflecting a healthy completion culture

1. Top Performing Genres by Audience Rating

This analysis ranks all comic genres by their mean audience rating (out of 10) to identify which genre combinations generate the strongest viewer engagement.



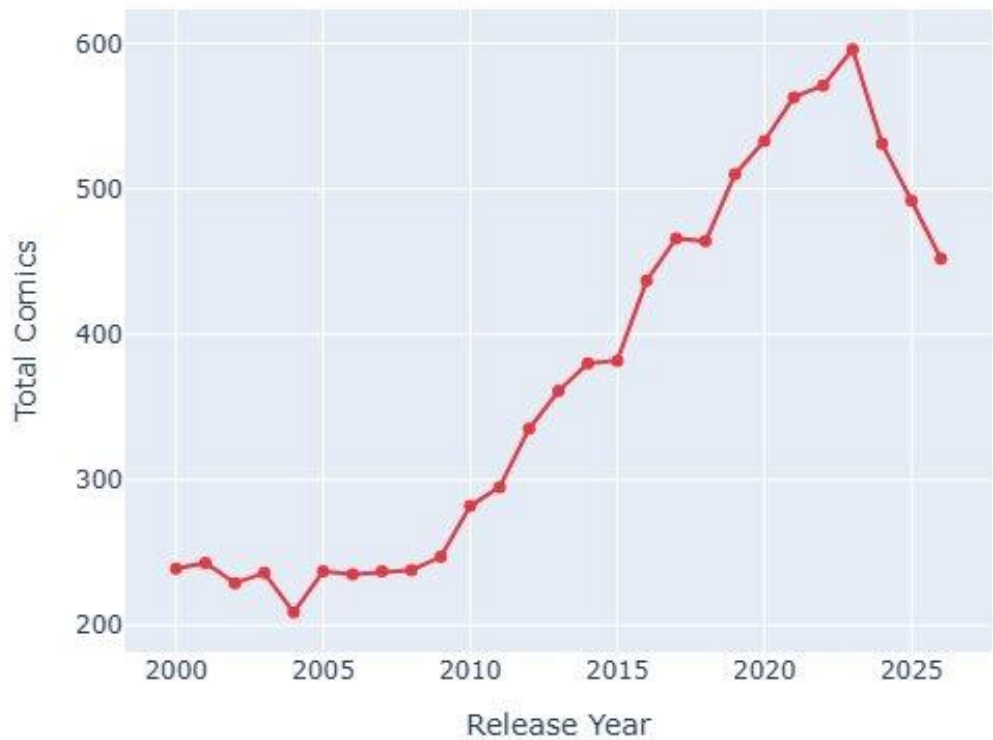
Key Insights

#	Insight
1	Adventure/Action leads all genres with a near-perfect rating close to 9.8/10, suggesting audiences strongly favour high-energy, escapist content.
2	The top 4 spots are all Action-blended genres (Adventure/Action, Historical/Action, Action/Mystery, Action/Sci-Fi), confirming that pairing Action with any secondary genre is a reliable formula for high audience satisfaction.
3	Ratings fall noticeably from ~9.3 (top 4) to ~8.3 at rank 5 (Action/Supernatural), revealing a two-tier structure — a high-performing elite cluster and a large mid-tier pack.
4	From rank 5 onward, roughly 20+ genres share ratings between 8.0 and 8.3 — a spread of just 0.3 points. Audience perception of these genres is nearly interchangeable.
5	Genres like Autobiography/Drama, Fantasy/Drama, and Literary/Drama anchor the bottom at ~7.8–8.0, reflecting a narrower, more critical audience base.

2. Comic Production Trend Over Time

A time-series view of annual comic production volumes from 2000 to 2026, capturing the industry's growth arc, peak, and recent correction.

Comic Production Trend Over Time



Key Insights

#	Insight
1	Production remained flat between ~210–245 comics annually from 2000 to 2008, suggesting a consolidation phase with limited appetite for expanding output.
2	Output nearly tripled — from ~245 to a peak of ~595 — over just 14 years (2009–2023), driven by the global boom in comic adaptations, streaming demand, and the rise of webtoon and digital publishing platforms.
3	Production hit its highest recorded point at approximately 595 comics in 2023, setting a benchmark no prior year came close to matching.
4	Output dropped steeply from ~595 (2023) to ~450 (2026) — a fall of roughly 145 units in just three years, likely signalling market saturation or a post-pandemic content correction.
5	At ~450 comics, 2026 production is back to 2015 levels. Whether this is a temporary correction or the start of a structural decline is the key strategic question for the industry.

3. Ratings Across Age Categories

A funnel chart comparing average audience ratings across the five official age classification categories to assess whether content maturity level influences perceived quality.

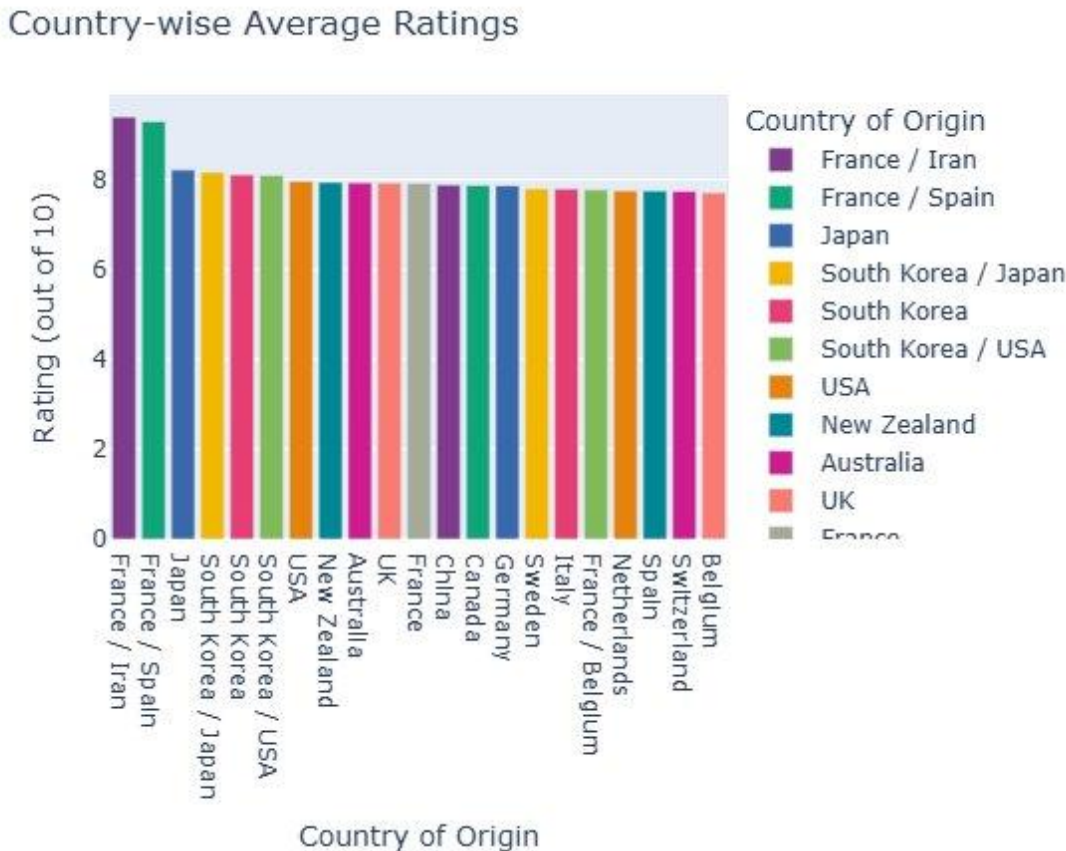


Key Insights

#	Insight
1	Young Adult tops the chart at 8.078/10, but the gap between first and last place (Mature at 8.040) is just 0.038 points — age rating has virtually no meaningful impact on audience scores.
2	All Ages content ranks second at 8.075, nearly matching Young Adult, suggesting universally accessible comics are crafted with consistently high quality.
3	Mature (8.040) and Mature 17+ (8.051) rank at the bottom two positions, challenging the assumption that adult-oriented content automatically earns more critical appreciation.
4	All five age categories sit within an extraordinarily tight range of 8.04 to 8.08. Story quality, art, and execution — not age classification — are the true drivers of audience satisfaction.
5	Teen+ earns 8.062, solidly in the middle, benefiting from a large, highly engaged readership base and genre diversity (action, fantasy, romance).

4. Country-wise Average Ratings

Comparison of average audience ratings by country (or collaborative country pair) of origin, identifying which national comic traditions resonate most strongly with global audiences.



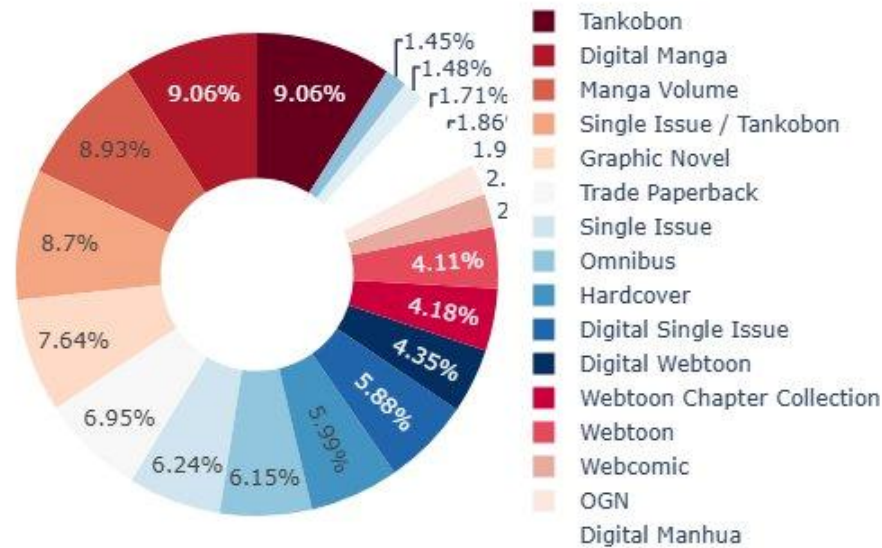
Key Insights

#	Insight
1	France/Iran and France/Spain claim the top two spots with ratings of ~9.1 and ~9.0 respectively, driven by auteur-style storytelling traditions that resonate strongly with global audiences.
2	Japan, South Korea/Japan, South Korea, and South Korea/USA all cluster between 8.1–8.2, confirming that East Asian comic industries consistently deliver high-quality content.
3	USA, New Zealand, Australia, UK, France, China, Canada, Germany, and Sweden all sit within a narrow 7.9–8.0 band, pointing to a global quality baseline for comics.
4	Belgium, Switzerland, Spain, Netherlands, and France/Belgium anchor the bottom at ~7.7–7.8 — still strong scores indicating even the lowest-ranked origins produce genuinely appreciated content.
5	Collaborative origin tags (France/Iran, France/Spain, South Korea/Japan, South Korea/USA) consistently outperform solo-nation entries, implying cross-cultural partnerships produce a measurable lift in quality perception.

5. Comic Format Popularity

A donut chart breaking down the relative market share of all comic delivery formats, from traditional print to digital-first and platform-native formats.

Comic Format Popularity

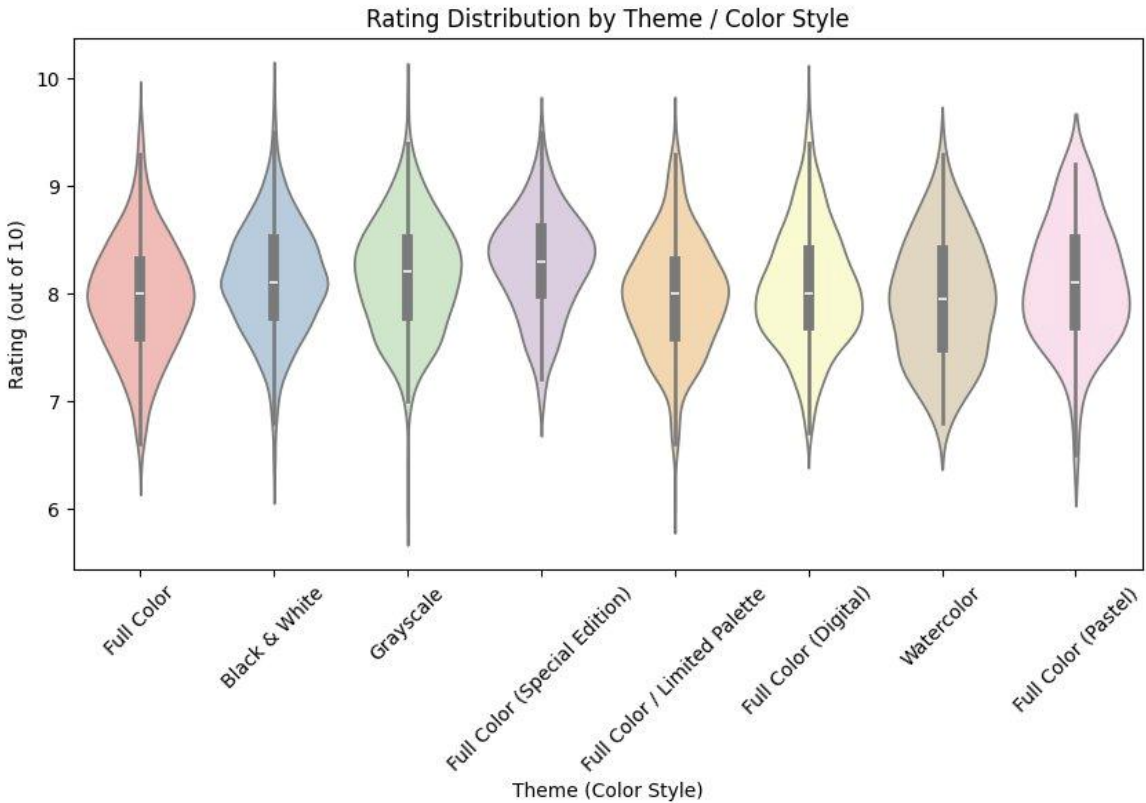


Key Insights

#	Insight
1	Tankobon (9.06%), Digital Manga (9.06%), and Manga Volume (8.93%) together account for over 27% of all comic formats, making Japanese-origin formats the single largest format bloc and reflecting manga's overwhelming global reach.
2	The largest individual share is just 9.06%, shared by Tankobon and Digital Manga. This near-equal fragmentation across 16+ formats indicates a highly diversified market.
3	Digital Manga (9.06%), Digital Single Issue (5.88%), Digital Webtoon (4.35%), and Digital Manhwa together represent a substantial combined share, signalling digital distribution is mainstream, not niche.
4	Webtoon (4.11%), Webtoon Chapter Collection (4.18%), and Digital Webtoon (4.35%) collectively account for ~12.6% of the market, reflecting the rapid global rise of the vertical-scroll format.
5	Graphic Novel (7.64%), Trade Paperback (6.95%), Single Issue (6.24%), and Hardcover (5.99%) still hold a combined share of over 26%, demonstrating that physical print formats retain strong reader demand.

6. Rating Distribution by Theme / Color Style

Violin plots showing the full distribution of audience ratings segmented by visual color style, assessing whether artistic style influences audience satisfaction.

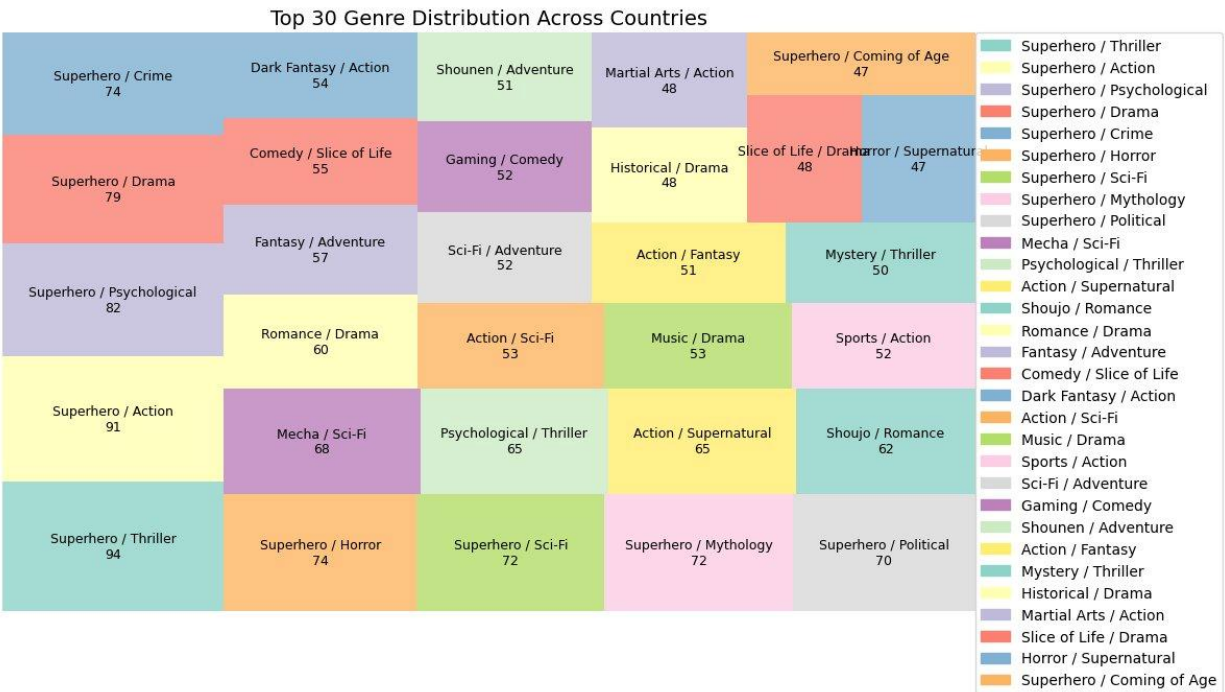


Key Insights

#	Insight
1	Across all eight color styles, the median rating sits consistently at approximately 8.0/10, confirming that color style alone does not meaningfully influence the central tendency of audience ratings.
2	Black & White extends notably toward 10.0 with a broad upper body, producing the highest concentration of top-tier rated titles. Classic manga and graphic novel traditions in B&W likely anchor this strong upper-end performance.
3	Full Color / Limited Palette has the narrowest and most pinched violin shape, indicating the least variance in ratings — predictable in quality, rarely exceptional or poor.
4	Full Color (Digital) extends furthest downward toward ~6.0, suggesting greater quality inconsistency. Democratisation of digital tools enables more creators to publish but with uneven results.
5	Watercolor and Full Color (Pastel) styles produce compact, symmetrical violin shapes with IQRs tightly packed around 7.8–8.2, reflecting calibrated audience expectations for these niche artistic styles.

7. Top 30 Genre Distribution Across Countries

A treemap visualising the top 30 genre-country combinations by count, revealing which genre archetypes have achieved the broadest international distribution.

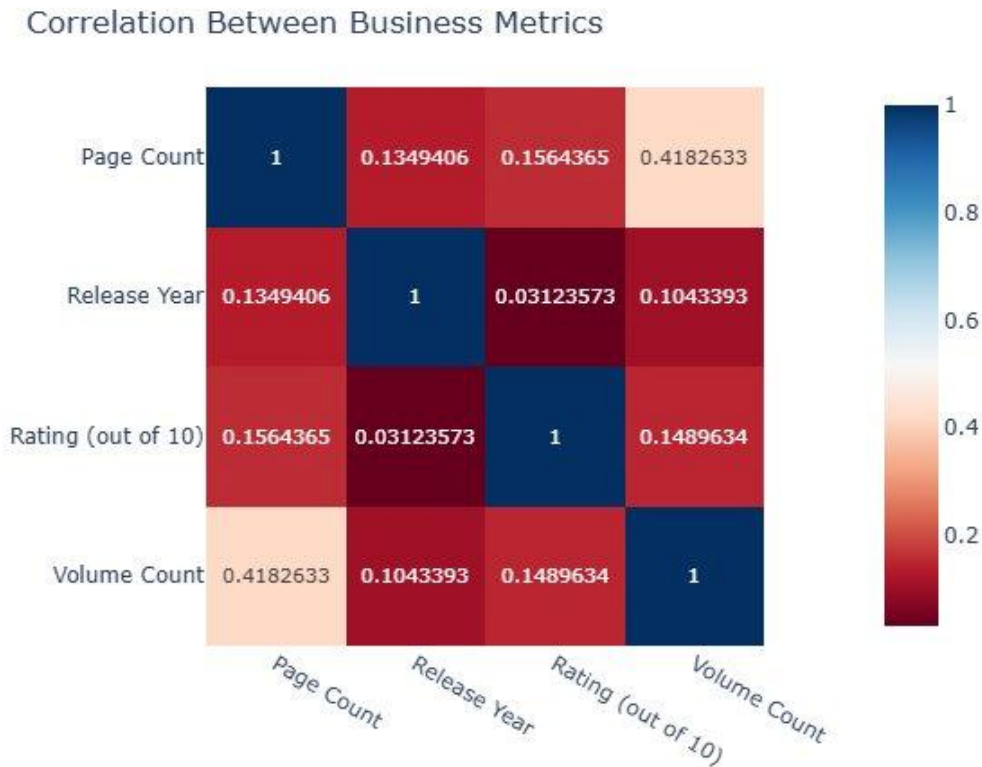


Key Insights

#	Insight
1	Superhero/Thriller leads at 94 — the single most globally distributed genre, confirming superhero content with dark, suspenseful undertones has the broadest cross-country appeal.
2	Superhero/Action follows closely at 91, confirming that action remains the most universally marketable pairing for superhero narratives across diverse cultural markets.
3	Superhero genres dominate 8 of the top 30 slots, reflecting the genre's unmatched global franchise infrastructure and consistent international audience demand.
4	Superhero/Psychological at 82 outranks pure action blends, showing audiences worldwide have a strong appetite for mentally complex, character-driven superhero storytelling.
5	Non-superhero genres cap out at 68 (Mecha/Sci-Fi), highlighting a clear ceiling for niche genres versus the mainstream dominance of the superhero category globally.

8. Correlation Between Business Metrics

A correlation heatmap examining the pairwise relationships between four core business dimensions: Page Count, Release Year, Rating, and Volume Count.

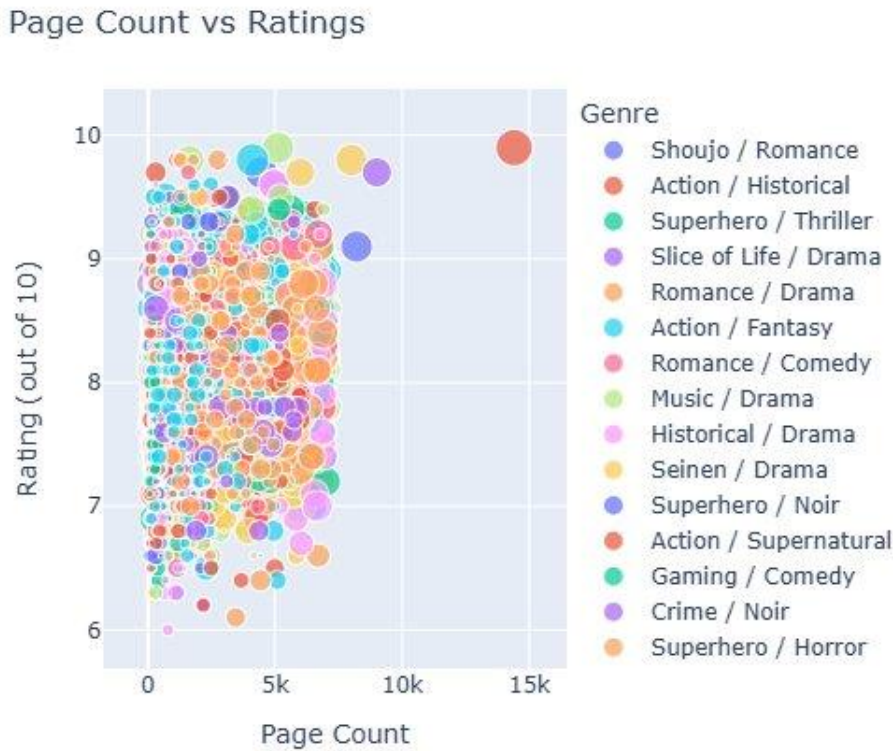


Key Insights

#	Insight
1	Page Count & Volume Count share the strongest relationship (0.42): longer comics tend to span more volumes, confirming that content depth and series length scale together as a natural publishing pattern.
2	Release Year is virtually uncorrelated with everything — its highest correlation is just 0.13 with Page Count, meaning newer comics are neither longer, better rated, nor more voluminous than older ones.
3	Rating is statistically independent of all other metrics: correlations of 0.16, 0.03, and 0.15 confirm that page count, recency, and volume count have no meaningful influence on audience scores.
4	Release Year & Rating produce the weakest link in the matrix (0.03), definitively ruling out recency bias — audiences do not rate newer comics higher simply because they are new.
5	No strong correlations exist anywhere in this matrix (maximum 0.42), signalling that comic success is driven by qualitative factors outside these four measurable dimensions.

9. Page Count vs. Ratings

A bubble scatter chart plotting individual comics by page count (x-axis) against audience rating (y-axis), with bubble size representing volume count and colour indicating genre.

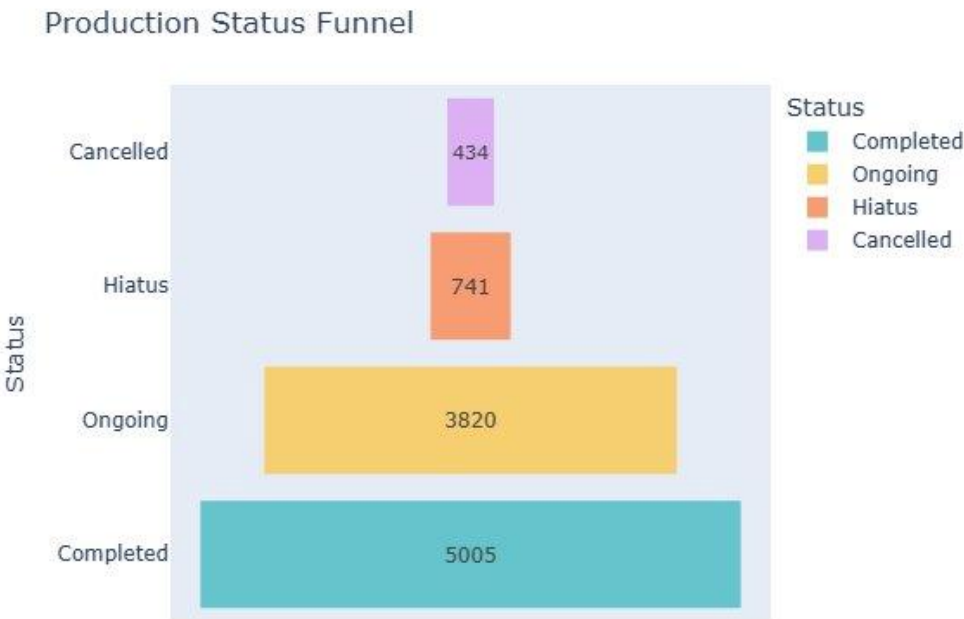


Key Insights

#	Insight
1	Top-rated titles (9–10) appear across the full 0–15k page range, proving that length has no bearing on audience satisfaction — quality is format-agnostic.
2	The overwhelming majority of titles cluster below 5k pages, with long-running series above 10k pages being rare outliers, confirming that most comics are relatively short-form.
3	A single Action/Sci-Fi title dominates at ~15k pages with a perfect 10 rating — the longest and simultaneously the highest-rated comic in the dataset, reflecting a uniquely dedicated fanbase.
4	Titles beyond 8k pages rarely dip below 8.0, implying that long-running series self-select for quality since only consistently well-received comics sustain readership long enough to accumulate high page counts.
5	Genre diversity is uniform across all page count ranges — no single genre monopolises long-form or short-form territory, with all major genres represented across the full length spectrum.

10. Production Status Funnel

A funnel chart breaking down the current production status of all titles in the dataset: Completed, Ongoing, Hiatus, and Cancelled.



Key Insights

#	Insight
1	Completed titles dominate at 5,005 — over 49% of all comics have reached completion, indicating a mature, back-catalogue-rich industry where readers can experience finished narrative arcs.
2	Ongoing titles at 3,820 represent a healthy active pipeline — nearly 38% of comics are still in production, signalling strong industry output momentum and large volumes of content actively retaining readership.
3	Hiatus titles at 741 are a significant retention risk — 7.3% of comics are stalled, placing thousands of readers in limbo whose engagement and platform loyalty are at risk the longer titles remain inactive.
4	Cancelled titles at 434 are the smallest but most permanent loss — at just 4.3% of total, representing a complete and unrecoverable audience investment with no future resolution for readers.
5	The Completed-to-Cancelled ratio of 11.5:1 signals a healthy completion culture — for every cancelled title, over 11 reach proper completion, reflecting a publishing ecosystem where started projects are seen through to their conclusion.

Strategic Recommendations

Based on the full EDA, the following strategic directions are recommended for publishers, platform operators, and content investors:

Strategic Priority	Recommended Action
Genre Strategy	Double down on Action-hybrid genre pairings. Greenlighting Adventure/Action, Historical/Action, and Action/Mystery projects delivers the highest probability of top-decile audience ratings.
Production Planning	Investigate the 2024–2026 output decline urgently. If driven by platform investment cuts, rebalancing toward mid-budget, high-frequency titles may stabilise volumes without overextending resources.
International Co-Productions	Franco-international and East Asian collaborative projects consistently outperform solo-nation entries. Structured co-production programmes between France, Japan, and South Korea should be prioritised.
Format Portfolio	Invest in the webtoon ecosystem (~12.6% combined share) as the highest-growth format channel. Simultaneously maintain print presence, which still commands 26%+ combined share.
Hiatus Management	Address the 741 hiatus titles as a retention priority. A structured 'revival fund' or creative partnership programme could reactivate stalled series before readership fully disengages.
Quality Over Length	Rating is independent of page count, volume count, and recency. Invest in editorial quality, world-building, and character development rather than volume or length targets.

Methodology & Data Notes

The analysis was performed on the Comic Books Dataset (CSV format) using Python and the following libraries:

- Pandas — data loading, cleaning, and aggregation
- Plotly Express — interactive bar charts, scatter plots, funnel charts, and heatmaps
- Seaborn — violin plots and static statistical visualisations
- Matplotlib — figure management and treemap rendering (Squarify)
- SciPy — skewness, kurtosis, and z-score outlier detection

Data preprocessing included: dropping null rows, type inference validation, and outlier detection using both the z-score method ($|z| > 3$) and the IQR method ($1.5\times$ IQR fence). Time-series aggregation was performed at the annual level. All ratings are on a 0–10 scale. All visualisations reflect post-cleaning data.

End of Report